PAID TECHNICAL ASSESSMENT

Founding AI Software Engineer - Engineering AI QA Assistant Mini MVP

Candidate	Raymond Balagtas
Assessment fee	AUD 200
Timeframe	3-5 days from receipt of the complete test pack
Purpose	Technical assessment only - not production work

1. Assessment Objective

Build a small prototype that demonstrates how you would design an AI-assisted engineering QA workflow for reviewing project documents, answering engineering questions with traceable evidence, managing revisions, and escalating uncertainty for human review.

> The prototype does not need to be production-ready. We are assessing your engineering judgment, AI/RAG architecture, document handling, traceability, safety, workflow design, and ability to explain technical decisions.

2. Test Pack

You will receive a small sample engineering project pack containing:

• 1 engineering drawing PDF

• 1 engineering design/report PDF

• 1 standards or internal QA checklist

• 1 revised version of a document

• 1 intentionally conflicting, incomplete, or missing-information scenario

3. Required Prototype Capabilities

1. Document ingestion and metadata. Ingest the supplied files and retain useful metadata such as filename, document type, revision, page, section, date/status where available.

2. Engineering Q&A; with citations. Allow a user to ask questions about the project and return answers supported by source references. At minimum, show the source file and page; section or other evidence location is preferred where available.

3. Revision control. Distinguish between revisions and demonstrate a method for prioritising the latest approved/current information rather than blindly using an older document.

4. Conflict and missing-information handling. Detect when the supplied documents conflict or when the answer cannot be verified. The system must not invent an answer when evidence is missing.

5. Human-in-the-loop QA. Include a simple review workflow such as: AI Flagged -> Engineer Review -> Accept / Reject / Needs Correction.

6. Basic user interface. Provide a simple interface for document upload or project selection, questions/findings, citations, and review status. A polished production UI is not required.

7. Audit trail. Keep a simple record of AI findings, evidence used, and reviewer actions/decisions.

4. Built-in Evaluation Scenarios

The test pack will intentionally include scenarios designed to test the safety and reliability of your approach:

• **Revision trap:** an older revision may contain a plausible answer that should not override the latest/current document.

• **Missing-answer trap:** at least one question will not be fully answerable from the supplied documents. The expected behaviour is to state that the answer cannot be verified and identify what information is missing.

• **Conflict scenario:** if two documents disagree, surface the conflict and the relevant sources rather than silently selecting an answer without explanation.

5. Deliverables

• Working prototype or local demo

• Source code repository, e.g. GitHub or equivalent

• README with setup and run instructions

• Brief architecture diagram or architecture overview

• 5-10 minute walkthrough video or recorded demo

• Short explanation of your approach to RAG, revision handling, hallucination control, human approval, security, and what you would improve for production

6. Technical Freedom

You may choose the language, framework, vector database, embedding model, LLM provider, OCR/document parser, and frontend approach you consider appropriate. Use of open-source or hosted services is acceptable. Please document any external services, API keys, or assumptions required to run the prototype.

7. Important Constraints

• Do not assume that every document is correct or current.

• Do not allow the model to present unsupported engineering statements as verified facts.

• Citations and traceability are required for engineering answers/findings.

• The prototype should make uncertainty and missing information visible to the reviewer.

• Human engineering review must remain part of the workflow for flagged or consequential findings.

• Do not expose confidential test materials publicly. If using a public repository, do not upload the supplied project documents.

8. Scoring Criteria

Criterion	Weight
Document ingestion & metadata	10%
Source citations / traceability	15%
Revision control	15%
Missing/conflicting information handling	15%
AI / RAG approach	15%
Human review & safety workflow	10%
UI / usability	5%
Code quality & architecture	10%
Technical explanation / walkthrough	5%

9. Submission

Please submit your completed assessment within 3-5 days of receiving the complete test pack. Send the repository link, README, architecture overview, demo/video link, and any access instructions required to review the prototype.

If you encounter an ambiguity in the test pack, you may document your assumption or ask for clarification. We are interested in how you handle incomplete information, so identifying uncertainty is preferable to guessing.

10. Payment & Use of Work

The assessment fee is AUD 200, payable upon completion and submission of the requested deliverables. This is a paid recruitment assessment. The prototype is being requested for evaluation purposes only and is not expected to be production-ready. Any future production development, commercial deployment, or additional work would be discussed separately.

11. What We Are Looking For

We are not looking for the largest or most complex build. We are looking for a clear, reliable and well-reasoned solution that demonstrates strong AI engineering fundamentals, good document/revision discipline, transparent evidence handling, and practical judgment for an engineering environment.